MEP Monthly Report

Technical Guide

Elephant sightings, patrols, collar voltages, and sitrep analysis

Generated August 31, 2026

Workflow id: **monthly_report**

1. Overview

The **monthly_report** workflow fetches data from EarthRanger to produce the MEP (Mara Elephant Project) Monthly Report. It covers all events (scatter map coloured by event type), collar GPS speedmaps, vehicle and foot patrol trajectories, per-subject collar voltage charts, and a situation report (sitrep). All outputs are assembled into a populated Word document.

The workflow delivers:

- 1 events scatter map (all event types, coloured by type via tab20 palette) + 1 events CSV export
- 1 elephant GPS speedmap
- 1 vehicle patrol trajectories map
- 1 foot patrol trajectories map
- Per-subject historic voltage charts — current voltage vs. a previous-period min/mean/max band
- 1 sitrep report — CSV export plus a sortable/filterable dashboard table
- An interactive dashboard — 6 widgets: Collar Voltage, Elephant Sightings Map, Speed Map, Vehicle Patrols Map, Foot Patrols Map, Sitrep Report
- A Word document report (overall_report.docx) — every map, chart, and the sitrep table
- A separate Word cover page — MEP logo (auto-downloaded), report period, prepared-by

Output summary

Output type	Count	Description
Events CSV	1	All fetched events with event_details flattened into columns
Events scatter map	1	Scatter layer of all events, coloured by event type (tab20)
Speedmap	1	GPS trajectories coloured by 6-bin speed classification
Vehicle patrol map	1	Vehicle patrol trajectories coloured by team (viridis)
Foot patrol map	1	Foot patrol trajectories coloured by team (viridis)
Vehicle patrol GeoParquet	1	Raw vehicle patrol trajectory data
Foot patrol GeoParquet	1	Raw foot patrol trajectory data
Relocations GeoParquet	2	Current-period and previous-period relocations, each with extracted voltage
Historic voltage charts	1 per subject	Current voltage vs. previous-period min/mean/max band
Sitrep report	1	CSV export + sortable/filterable dashboard table
Dashboard widgets	6	Collar Voltage, Sightings, Speed, Vehicle, Foot, Sitrep
Word report	1	Content report (overall_report.docx), all maps/charts/sitrep
Word cover page	1	Separate document — MEP logo, report period, prepared-by

2. Dependencies

2.1 Python packages

Package	Version	Channel
ecoscope-platform	>=2.15.0, <2.16.0	ecoscope-workflows
ecoscope-workflows-ext-custom	0.1.0rc14.*	ecoscope-workflows-custom
ecoscope-workflows-ext-ste	0.0.0rc1.*	ecoscope-workflows-custom
ecoscope-workflows-ext-mep	1.0.1.*	ecoscope-workflows-custom
pydeck	0.9.2	conda-forge
opentelemetry-sdk	>=1.20.0, <2.0.0	conda-forge

2.2 Connections

Connection	Task	Purpose
EarthRanger	set_er_connection	Fetch events, subject group observations, vehicle and foot patrol data, and sitrep events

2.3 Dropbox files

Two Word template files are downloaded from Dropbox at runtime if not already present (**overwrite_existing: false**, retries: 3):

File	Purpose	Dropbox URL (abbreviated)
mep_monthly_report.docx	Word template for the cover page (title, report period, prepared-by field)	dropbox.com/…/mep_monthly_report.docx?rlkey=nbibg8ulnlz0w4q53jw2db6y3
mep_monthly_indv_report.docx	Word template for the content page (maps, charts, collar charts, sitrep table)	dropbox.com/…/mep_monthly_indv_report.docx?rlkey=wss0x8sa9i5fgl9yjco7paa03

2.4 Base maps

Base map tiles are configured by the user via **set_base_maps_pydeck**. Any tile layer supported by deck.gl can be specified.

2.5 Grouper

The workflow groups all data by **name** (subject name). The grouper is fixed to **index_name: name** — users cannot change the grouping dimension.

2.6 Subject group

A user-provided string parameter (**Subject Group Name**) is passed to both the current-period and previous-period observation fetches, and also included in the Word report content page context. Every subject in the group receives a collar voltage chart.

3. Events Pipeline

All events within the analysis time range are fetched, normalized, coloured by event type, and rendered as a scatter map.

3.1 Event retrieval, normalization, and colourmap

Step	Task	Detail
1	get_events	Fetch all events for the analysis time range (no event_type filter). Columns retained: id, time, event_type, event_category, reported_by, serial_number, geometry, created_at, event_details. include_details: true, include_null_geometry: false, force_point_geometry: true, raise_on_empty: true.
2	normalize_json_column	Flatten the event_details JSON column into top-level columns (skip_if_not_exists: true, sort_columns: true).
3	persist_df	Persist the normalized events as events.csv to ECOSCOPE_WORKFLOWS_RESULTS.
4	apply_color_map	Apply the tab20 palette to the event_type column, writing colours to the output column event_type_colors. Each distinct event type receives a unique colour from the tab20 palette.

Note: Spatial-outlier exclusion and null-geometry dropping (previously exclude_geom_outliers / drop_null_geometry) are no longer part of this pipeline — events are fetched with force_point_geometry: true and mapped directly.

3.2 Events scatter map

Parameter	Value
Layer type	ecoscope_workflows_ext_custom.tasks.results.create_scatterplot_layer
Fill color	event_type_colors (tab20, dynamic per event type)
Line color	event_type_colors (same column)
Line width	0.55
Radius	3.55 m
Opacity	0.55
Stroked	true
Legend title	Legend
Legend label column	event_type
Legend color column	event_type_colors
View state	compute_view_state_from_gdf (ext_ste), max_zoom: 15
Draw task	ecoscope_workflows_ext_custom.tasks.results.draw_map

Parameter	Value
Max zoom	10
Screenshot timeout	40 000 ms (map tile rendering)

The map HTML is persisted as **elephant_sightings_map.html** then converted to PNG with device_scale_factor: 2.0.

4. Subject GPS Speedmap Pipeline

Collar GPS relocations for the subject group are converted to trajectories and coloured by speed to produce an overall speedmap.

4.1 Observations and relocations

Step	Task	Detail
1	get_subjectgroup_observations	Fetch observations for the current time range (filter: clean, include_details: true, include_subjectsource_details: true, raise_on_empty: false).
2	get_timezone_from_time_range → convert_values_to_timezone	Convert the fixtime column to the analysis timezone.
3	process_relocations	Retain 12 columns: groupby_col, fixtime, junk_status, geometry, extra__subject__name, extra__subject__hex, extra__subject__sex, extra__created_at, extra__subject__subject_subtype, extra__subjectsource__id, extra__subjectsource__assigned_range, extra__observation_details. Filter 3 invalid coordinate pairs: (180,90), (0,0), (1,1).
4	sort_values → persist_df	Sort by fixtime ascending, then persist as GeoParquet (relocations.parquet). This sorted GeoDataFrame feeds both the trajectory/speedmap pipeline below and the historic voltage pipeline in Section 8.

4.2 Previous period observations

A parallel fetch retrieves observations for a user-configurable **Previous Period** (via ecoscope_workflows_ext_ste.tasks.filter.flexible_previous_period — Custom offset, a Preset lookback, or an exact Calendar date), processed identically (timezone conversion, process_relocations, sort, persist as previous_period_relocations.parquet). See Section 8 for how the current and previous relocation sets are combined into the historic voltage charts.

4.3 Trajectory segment filter

Parameter	Value	Description
min_length_meters	0.001	Minimum segment length
max_length_meters	5 000	Maximum segment length
min_time_secs	1	Minimum time between fixes
max_time_secs	21 600	Maximum time (~6 hours)
min_speed_kmhr	0.01	Minimum plausible speed
max_speed_kmhr	9	Maximum plausible speed

4.4 Speed classification and colormap

After adding a temporal index (time_col: segment_start), speed is classified into **6 equal-interval bins** with label_ranges: true, label_decimals: 1, label_suffix: ' km/h'. Trajectories are sorted ascending by speed bin then coloured using the following ramp:

Bin	Hex color	Description
1 (slowest)	#1a9850	Dark green
2	#91cf60	Light green
3	#d9ef8b	Yellow-green
4	#fee08b	Light amber
5	#fc8d59	Orange
6 (fastest)	#d73027	Red

4.5 Speedmap layer and map

Parameter	Value
Column filter	subset_columns (was: filter_df_cols), exclude: null, strict: false
Layer type	ecoscope_workflows_ext_custom.tasks.results.create_path_layer
Color column	speed_bins_colormap
Width	2.85, width_units: pixels, min 2 / max 8 px
Cap / joint	rounded, billboard: false
Opacity	0.55
Legend	Speed (km/h), sorted ascending
View state	compute_view_state_from_gdf (ext_ste), max_zoom: 15
Draw task	ecoscope_workflows_ext_custom.tasks.results.draw_map
Max zoom	10
Screenshot timeout	40 000 ms

Columns retained for the map layer: dist_meters, speed_bins_colormap, geometry, speed_kmhr, speed_bins. The HTML is persisted as **elephant_speedmap.html** (filename_suffix: null) then converted to PNG.

5. Vehicle Patrol Pipeline

Vehicle patrol observations are fetched from EarthRanger, converted to trajectories, and rendered as a map coloured by patrol team.

5.1 Patrol retrieval

Task: **get_patrol_observations** with parameters: include_patrol_details: true, raise_on_empty: true, sub_page_size: 100, patrols_overlap_daterange: true. No patrol_type filter is applied — all vehicle patrol types present in EarthRanger for the time range are included.

5.2 Relocations

Task: **process_relocations**. Columns retained: patrol_id, patrol_start_time, patrol_end_time, geometry, patrol_type__value, patrol_type__display, patrol_serial_number, patrol_status, patrol_subject, groupby_col, fixtime, junk_status, extra__source. Invalid coordinate pairs (180,90), (0,0), (1,1) are filtered.

5.3 Trajectory segment filter

Parameter	Value
min_length_meters	0.35
max_length_meters	5 000
min_time_secs	1
max_time_secs	18 000 (~5 hours)
min_speed_kmhr	10
max_speed_kmhr	100

Note: The minimum speed of 10 km/h filters out stationary periods and slow foot movement, retaining only motorised patrol segments.

5.4 Colormap and map layer

Trajectories are coloured by **extra__patrol_type__value** using the **viridis** colormap (task: apply_color_map, output_column: patrol_type_colormap). A path layer is created via **ecoscope_workflows_ext_custom.tasks.results.create_path_layer** with the same style as the speedmap (width 2.85, opacity 0.55, rounded). View state comes from **ecoscope_workflows_ext_ste.tasks.spatial_operations.compute_view_state_from_gdf** (max_zoom: 15), and the map is drawn via **ecoscope_workflows_ext_custom.tasks.results.draw_map**. Legend title: *Patrol team*, sorted ascending.

5.5 Persistence

Trajectories are persisted as **vehicle_patrol_trajectories.geoparquet**. The map HTML is persisted as **vehicle_patrols_map.html** (filename_suffix: null) then converted to PNG with a 40 000 ms screenshot timeout.

6. Foot Patrol Pipeline

The foot patrol pipeline mirrors the vehicle patrol pipeline but uses different trajectory thresholds appropriate for walking speeds. No patrol_type filter is applied — all foot patrol types present in EarthRanger for the time range are included.

6.1 Trajectory segment filter

Parameter	Value
min_length_meters	0.001
max_length_meters	5 000
min_time_secs	1
max_time_secs	14 400 (~4 hours)
min_speed_kmhr	0.5
max_speed_kmhr	9

6.2 Colormap, map layer, and persistence

Identical to the vehicle patrol pipeline (same renamespaced create_path_layer / draw_map / compute_view_state_from_gdf tasks): viridis colormap on extra__patrol_type__value, path layer with legend *Patrol team*. Trajectories persisted as **foot_patrol_trajectories.geoparquet**, map as **foot_patrols_map.html** (filename_suffix: null) → PNG (40 000 ms timeout).

7. Sitrep (Situation Report) Pipeline

The workflow compiles a situation report from EarthRanger events configured by a region column lookup.

7.1 Configuration and compilation

Step	Task	Detail
1	get_sitrep_event_config	Retrieves the sitrep event type configuration keyed by region_column: 'region'. Returns an event_details mapping used to categorise events by region.
2	compile_sitrep	Fetches and aggregates EarthRanger events for the analysis time range using the event_details configuration. Produces a summary DataFrame of incident counts and categories per region.
3	persist_df	Saves the sitrep DataFrame as **sitrep_report.csv** to ECOSCOPE_WORKFLOWS_RESULTS. Skipped if the result is empty or a dependency was skipped.

7.2 Dashboard table

Step	Task	Detail
1	draw_table	Render the sitrep DataFrame as a sortable, filterable table (enable_sorting: true, enable_filtering: true, enable_download: false, widget_id: 'Sitrep Report').
2	persist_text	Persist the rendered table HTML as sitrep_report_table.html.
3	create_table_widget_single_view	Wrap the table HTML as the 'Sitrep Report' dashboard widget.

Note: The sitrep CSV (sitrep_report.csv) is picked up by filename when create_mep_monthly_report scans ECOSCOPE_WORKFLOWS_RESULTS for inclusion in the Word report's content — see Section 9.2 — while the sitrep table above feeds the dashboard independently.

8. Collar Voltage Charts

The single compound task previously used here (process_collar_voltage_charts) has been replaced by an explicit, reusable per-step pipeline shared with the standalone collar/source voltage workflows, built on the sorted, timezone-converted current and previous relocation GeoDataFrames from Section 4.

8.1 Voltage extraction

Both relocation sets go through **extract_value_from_json_column** against their **observation_details** column, producing a float **voltage** column:

Parameter	Value
column_name	observation_details
field_name_options	battery, mainVoltage, batt, power (checked in this order)
output_type	float
output_column_name	voltage

8.2 Per-subject fan-out

Step	Task	Detail
1	split_groups	Split current and previous relocations by the configured grouper (name)
2	column_first_unique_value	Get each subject's display name from the current-period slice
3	safe_string (ext_ste)	Sanitize the subject name for use in a filename
4	prefix_string_var	Build the chart filename: _historic_voltage.html
5	groupbykey	Pair each subject's current-period slice with its previous-period slice (zip_current_prev_name)

8.3 Plotting

plot_historic_voltage (column: voltage) draws the subject's current voltage series against a band built from the previous period's 2.5th/97.5th percentile and mean. If the previous-period slice is missing or empty, the current period's own values are used for the band instead. If the computed band collapses to a single value, it is widened by ±2.5% so the shaded region remains visible.

8.4 Persisting, widget, and PNG

Step	Task	Detail
1	groupbykey	Pair each chart's filename with its rendered HTML (historic_voltage_text); skipped if any dependency was skipped or any keyed pair is a skip (any_keyed_iterables_are_skips, unpack_depth: 1)

Step	Task	Detail
2	persist_text	Write the chart HTML to ECOSCOPE_WORKFLOWS_RESULTS (filename_suffix: null — the filename built in 8.2 is used verbatim)
3	create_map_widget_single_view	Title: 'Collar Voltage'; skipif: never, so a widget is always created even if the underlying chart data is empty
4	merge_widget_views	Merge every subject's widget into a single dashboard widget
5	html_to_png	device_scale_factor: 2.0, wait_for_timeout: 10 ms, max_concurrent_pages: 1, full_page: false

9. Word Report

The cover page and the content report are now two separate final documents — there is no longer a merge step producing a single combined file.

9.1 Cover page

Step	Task	Detail
1	fetch_and_persist_file	Download mep_monthly_report.docx (cover template) from Dropbox.
2	ecoscope_workflows_ext_ste.tasks.io.fetch_and_persist_file	Download the MEP organisation logo (MEP-logo-dark-linear.png) from Dropbox — always applied, not user-configurable (overwrite_existing: false, retries: 2).
3	prepare_cover_metadata	Build the cover context: org_logo_path (from step 2), report_period (analysis time range), prepared_by: 'Ecoscope', extra_fields: null, time_generated_format: '%Y-%m-%d %H:%M:%S'.
4	create_context_page	Populate the cover template with the context. skipif: any_dependency_skipped (unpack_depth: 1). Output filename: mep_monthly_cover_page.docx.

9.2 Content report

Step	Task	Detail
1	fetch_and_persist_file	Download mep_monthly_indv_report.docx (content template) from Dropbox.
2	create_mep_monthly_report	Takes only template_path and output_dir — no explicit chart/CSV paths. It walks ECOSCOPE_WORKFLOWS_RESULTS and classifies files by their known filename stems: elephant_speedmap, elephant_sightings_map, vehicle_patrols_map, foot_patrols_map (single images), any _historic_voltage image (grouped by subject, suffix stripped), and sitrep_report.csv (rendered as the sitrep table). Output filename: overall_report.docx.

Note: Because the content report is assembled by scanning ECOSCOPE_WORKFLOWS_RESULTS for these specific filenames rather than receiving explicit paths, renaming any upstream persist_text / persist_df filename above will silently drop that chart or table from the report instead of raising an error.

10. Output Files

All outputs are written to the directory specified by **ECOSCOPE_WORKFLOWS_RESULTS**. Files marked with *<subject>* are produced once per collared subject.

File	Description
events.csv	All fetched events with event_details flattened into columns
elephant_sightings_map.html / .png	Scatter map of all events, coloured by event type (tab20)
elephant_speedmap.html / .png	GPS trajectories coloured by 6-bin speed classification
vehicle_patrols_map.html / .png	Vehicle patrol trajectories coloured by team (viridis)
foot_patrols_map.html / .png	Foot patrol trajectories coloured by team (viridis)
vehicle_patrol_trajectories.geoparquet	Raw vehicle patrol trajectory GeoDataFrame
foot_patrol_trajectories.geoparquet	Raw foot patrol trajectory GeoDataFrame
relocations.parquet	All subjects' current-period relocations, including extracted voltage
previous_period_relocations.parquet	All subjects' previous-period relocations, including extracted voltage
_historic_voltage.html / .png	Historic voltage chart — current voltage vs. previous-period min/mean/max band
sitrep_report.csv	Situation report: incident counts and categories by region
sitrep_report_table.html	Sitrep report rendered as a sortable/filterable table (dashboard widget source)
mep_monthly_cover_page.docx	Populated Word cover page (MEP logo, report period, prepared by)
overall_report.docx	Final Word monthly report (every map, chart, and the sitrep table)

11. Workflow Execution Logic

11.1 Global skip conditions

Every task now inherits the same skip conditions from a single top-level **task-instance-defaults** block, rather than each task declaring its own identical skipif (previously duplicated across roughly 30 individual task definitions):

Condition	Behaviour
any_is_empty_df	Skip task if any input DataFrame is empty
any_dependency_skipped	Skip task if any upstream dependency was skipped

Two tasks override this default:

- **collared_voltage_widget** uses **skipif: conditions: [never]**, so a dashboard widget is always created even if the underlying chart data is empty
- **persist_cover_page** and **historic_voltage_text** use **any_dependency_skipped** (the latter also **any_keyed_iterables_are_skips**, unpack_depth: 1), so an individual subject's filename/chart pair — or the cover page — is skipped without failing the whole zip

11.2 Screenshot timing

Map / Chart	wait_for_timeout	Reason
Events scatter map	40 000 ms	Tile map — waits for base tile rendering
Speedmap	40 000 ms	Tile map — waits for base tile rendering
Vehicle patrol map	40 000 ms	Tile map — waits for base tile rendering
Foot patrol map	40 000 ms	Tile map — waits for base tile rendering
Historic voltage charts	10 ms	Static Plotly HTML — no tiles

11.3 Dashboard

The workflow concludes with **gather_dashboard**, which packages workflow details, time range, groupers, and 6 widgets into the final interactive dashboard:

Widget	Source
Collar Voltage	grouped_collared_widget (merged per-subject historic voltage charts)
Elephant Sightings Map	sightings_map_widget (persist_sightings_urls)
Speed Map	speedmap_widget (persist_speedmap_html)
Vehicle Patrols Map	vehicle_map_widget (vehicle_patrol_map)
Foot Patrols Map	foot_map_widget (foot_patrol_map)
Sitrep Report	sitrep_table_widget (persist_sitrep_table_html)

Previously the **widgets** list was empty — this workflow had no interactive dashboard content beyond the Word report.

12. Software Versions

Package	Version pinned in spec.yaml
ecoscope-platform	>=2.15.0, <2.16.0
ecoscope-workflows-ext-custom	0.1.0rc14.*
ecoscope-workflows-ext-ste	0.0.0rc1.*
ecoscope-workflows-ext-mep	1.0.1.*
pydeck	0.9.2
opentelemetry-sdk	>=1.20.0, <2.0.0

Note: All packages are resolved from the prefix.dev Ecoscope conda channels. ecoscope-workflows-ext-mnc, ecoscope-workflows-ext-big-life, and ecoscope-workflows-ext-icf are no longer dependencies of this workflow. ecoscope-workflows-ext-ste is pinned to a pre-release (0.0.0rc1.*).